Step 2b - H7 - Ore weakening before comminution

Tech baseline

Technologies used in the market

Conventional crushing + SAG/ball milling (incumbent)

How it works: Multi-stage mechanical crushing followed by semi-autogenous and ball mills; breakage is indiscriminate, applied to the rock bulk rather than grain boundaries.

How well it performs: Total circuit ~9-23 kWh/t depending on ore hardness (Bond work index ~10-20 kWh/t); comminution = ~36-50% of mine-site energy; only ~3-5% of input energy creates new surface area.

Limits: Extremely inefficient; energy scales with ore competency; no mechanism to preferentially weaken ore — the hypothesis attacks exactly this inefficiency.

High-Pressure Grinding Rolls (HPGR)

How it works: Ore bed compressed between counter-rotating rolls; inter-particle crushing generates micro-cracks and a weaker, finer product feeding ball mills.

How well it performs: ~15-20% lower direct comminution energy vs SAG circuits (e.g. ~7.6-19.3 kWh/t vs 9-23 kWh/t for SABC); ~23-25% lower opex; pilot data shows up to ~60% machine-level energy reduction vs ball milling alone.

Limits: High capex, stud/roll wear, prefers dry feed, benefit is ore-dependent; still applies bulk mechanical stress rather than selective grain-boundary weakening — microwave can be complementary upstream, not a replacement.

High-voltage pulse pre-weakening (SELFRAG / electrodynamic fragmentation)

How it works: Sub-microsecond electrical discharges pass through immersed ore, generating plasma channels that fragment particles selectively along grain boundaries.

How well it performs: Pre-weakening at ~1-3 kWh/t; JKMRC/Newcrest simulation of a gold-copper circuit indicated ~5 kWh/t circuit energy reduction and potential removal of pebble crushers + a 10 MW ball mill.

Limits: Batch/low-throughput machines, requires water immersion and drying, electrode wear, limited industrial adoption at mining scale — microwave treats dry ore continuously at much higher throughput.

Mine-to-mill / blast optimisation (explosive pre-weakening)

How it works: Higher-intensity blasting and fragmentation modelling to deliver finer ROM feed, shifting size reduction from mills to cheaper explosive energy.

How well it performs: Typically ~5-15% comminution energy reduction achievable where applicable; explosive energy ~$0.1-0.3/t vs electricity ~10-20x costlier per equivalent breakage.

Limits: Diminishing returns, already adopted at most well-run sites, constrained by geotechnical/vibration limits — sets the bar a new pre-treatment must exceed.

Microwave pre-treatment (state of the art — the hypothesis technology)

How it works: High-power microwave energy selectively heats high-dielectric-loss mineral phases; differential thermal expansion generates inter- and trans-granular micro-cracks that

weaken the ore before it enters crushers/mills.

How well it performs: Nottingham/e2v pilot plant: 10-150 t/h at 10-200 kW, effective at ~0.3-3 kWh/t dose; Bond work index reduced ~3-9% (up to 30-40% in lab on amenable ores, e.g. Palabora 13.1→9.2 kWh/t in 10 s); flowsheet simulations: -9% to -24% specific comminution energy and up to +30% throughput; MicroHammer claims >20% energy reduction at 150 t/h demo, scalable to ~3,000 t/h.

Limits: Benefit highly ore/mineralogy-dependent (needs microwave-absorbing vs transparent phase contrast); unproven at full commercial scale; very short exposure (<0.1-1 s) needed for favourable net energy balance; no operating commercial installation yet.

Why our solution can be better

Microwave pre-treatment attacks the weakest point of the incumbent flowsheet: mills waste ~95% of their energy because ore must be stressed as a bulk solid, while microwaves deposit energy selectively at grain boundaries where breakage is most valuable, for ~0.5-2 kWh/t input. Unlike HPGR or new mills, it is an add-on installed between primary crushing and milling — it debottlenecks existing circuits (+~30% modelled throughput) and avoids multi-hundred-$M mill expansions, which is a far easier capex decision than circuit replacement. Versus its closest novel rival (HVP/electrodynamic), it treats ore dry, continuously, and has demonstrated 10-150 t/h pilots with a credible path to mining-scale throughput (e2v/Nottingham modelled ~3,000 t/h).

Key assumptions

• Comminution consumes ~36-50% of mine-site energy; the grinding mill is the largest single consumer (measured, Weir/Engeco 2021; US DOE bandwidth study: grinding ~40% of mining energy)

• Microwave doses of ~0.3-1.3 kWh/t at high power density measurably weaken ore at pilot scale: A*b -7-14%, Bond work index -3-9% (measured, Nottingham pilot, 300 t processed under power)

• Modelled -9% to -24% specific comminution energy and up to +30% throughput depending on ore and grind size (measured/simulated — not yet demonstrated at plant scale)

• Net energy balance is favourable: ~1 kWh/t microwave input vs ~2-5 kWh/t downstream saving; requires sub-second high-power-density exposure (assumed — simulation-based, ore-dependent)

• Sufficient fraction of world ores are amenable (need strong dielectric-loss contrast, e.g. sulphides/pyrite in quartz matrix); lab strength reductions range ~8-40% by ore (measured but highly variable — key risk)

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